LAB 2: Automated Car Wash & Detailing Bay Controller

Modular Functions, Infinite Loops, and Menu Systems

1. Introduction

1.1. Overview

In this lab, you will design and implement a modular C program that simulates an Automated Car Wash & Detailing Bay Controller. The system operates an interactive self-service kiosk where customers can purchase wash packages using a step-by-step coin payment mechanism, while system operators can access a password-protected admin mode to replenish supplies, adjust pricing, and track revenue.

The application must be structured using modular, single-responsibility functions and infinite control loops. Complete this assignment without using arrays, pointers, or C structures; track all system states (water, soap, wax inventory, package prices, and total revenue) using scalar variables and function parameter passing.

1.2. Student Learning Outcomes

- **1: Modular System Architecture** — translate system requirements into user-defined C functions.
- **2: Control Flow & Menu Systems** — Implement nested infinite control loops.
- **3: State & Resource Tracking** — Manage inventory (water, soap, wax) and sales revenue across functions using scalar parameters.
- **4: Input Validation & Loops** — Program robust coin payment processing (accepting 1.00, 0.50, and 0.25 AED) with exact change.
- **5: Debugging & State Tracing** — Trace execution flows and verify local variable states using VS Code breakpoints and watch windows.

2. Submission Guidelines

- File Format: Submit your C source code file (carwash_sim.c) and a lab report containing flowcharts and screenshots of different test case outputs.
- File Naming: Use the standard format: CSCI291_Lab2_StudentName_ID.c.
- Upload both files under the Lab 2 Submission Link on Moodle.
- Evaluation: Live in-lab code demonstration with the lab demonstrator and a lab report.

Lab session	Weekly Focus & In-Lab Goals	Key Deliverables
Week 2	**Architecture & Main Control Loop:** Set up `#define` constants, implement top-level infinite loop, display main menu, and code function stubs.	Pre-Lab Flowchart, Main Menu Code
Week 3	**Customer Mode & Payment Loop:** Code package selection, resource availability checking, step-by-step coin insertion (`processPayment`), change calculation, and alert triggers	`customerMode()` & `processPayment()`
Week 4	**Admin Operations & Debugging:** Code passcode check (`1357`), inventory display, manual replenishment, price editing, revenue reset, and live VS Code demonstration.	`adminMode()`, Final `carwash_sim.c`, full report

3. System specifications

The car wash controller manages three wash packages using specific ingredient consumption rates and pricing. The system operates from an initial inventory state and tracks cumulative sales.

3.1. Wash Package Specifications & Resource Requirements

The car wash offers three distinct service packages. Each wash cycle consumes precise quantities of water (Litres), soap concentrate (mL), and liquid wax (mL) from the machine's internal tanks. Table 3.1 details the ingredient requirements and default pricing for each package.

Table 3.1: Ingredient requirements and default pricing for each package.

Wash Package Name	Water (Litres)	Soap (mL)	Wax (mL)	Default Price (AED)
Basic Eco Wash	20	100	0	15
Deluxe Foam Wash	35	250	50	25
Premium Shine & Wax	50	300	150.	40

3.2 Inventory Tracking, Availability Checks & Low-Stock Alerts

System inventory is initialized at startup and tracked using individual scalar variables (e.g., water_level). Before offering a wash to a customer, the system calls isPackageAvailable() to compare remaining inventory against package requirements. If any ingredient is insufficient, the package is flagged as [UNAVAILABLE - Low Supplies]. Upon successful

order completion, ingredients are deducted immediately. If stock drops below minimum safety thresholds (Table 3.2), the system emits an operator warning alert.

Table 3.2: Safety thresholds for each resource

Resource Name	Initial Inventory	Low Supply Threshold (Alert)	Max Capacity
Water Supply	150 Liters	<= 30.0 Litres	300 Liters
Soap Concentrate	800 mL	<= 200 mL	2000 mL
Liquid Wax	400 mL	<= 50 mL	1000 mL

3.3 Coin Payment & Currency Validation

The kiosk accepts 1.00 AED, 0.50 AED, and 0.25 AED cash. If an unaccepted coin or invalid number is entered, the coin is rejected with a message, and the balance remains unchanged. Once total inserted coins meet or exceed the price, the system dispenses the wash, adds the transaction price to total_sales_revenue, and calculates change due: change = totalInserted - price.

3.4 Admin Security & Maintenance Mechanics

Access to adminMode() requires entering the passcode 1357. Upon authorization, operators can:

1. Inspect real-time supply levels and total cumulative sales revenue.
2. Manually Replenish Inventory: Prompts the operator to enter the specific quantity of Water (L), Soap (mL), and Wax (mL) to add. The code validates inputs and caps supply levels at maximum tank capacities (Water: 300 L, Soap: 2000 mL, Wax: 1000 mL).
4. Modify package unit prices in AED.
5. Reset total sales revenue to 0.00 AED after reminding the operator to collect cash box earnings.
6. Format currency to 2 decimal places and volumes to 1 decimal place.

4. Pre-Laboratory Tasks

Before writing code in VS Code, you must plan your program architecture visually.

- **Task 4.1: Function Call Hierarchy Chart:**
  Draw a block diagram showing how main() delegates execution to customerMode() and adminMode(), along with their respective sub-functions.
- **Task 4.2: Modular Control Flowchart:**
  Draw an ANSI flowchart for the processPayment() function detailing the coin insertion loop, coin validation checks (accepting 1.00, 0.50, 0.25 AED), balance updates, and change calculation.

5. Laboratory Tasks

Your source file (**carwash_sim.c**) must contain the following 11 required functions:

Function	Specification
int main(void)	Entry point. Runs the top-level `while(1)` loop, displays the main menu, and routes to Customer Mode, Admin Mode, or Exit.
void displayMainMenu(void)	Prints top-level interactive options (1: Purchase Wash, 2: Admin Operations, 3: Exit).
void customerMode(void)	Customer wash ordering module. Displays available packages, filters out packages if ingredients are depleted, prompts selection/confirmation, and calls payment processing.
int isPackageAvailable(int pkg)	Evaluates if current water, soap, and wax levels meet package minimums. Returns 1 (Available) or 0 (Unavailable).
void processPayment(float price, float reqWater, float reqSoap, float reqWax, int packageType)	Executes coin insertion loop (1.00, 0.50, 0.25 AED), rejects invalid coins, tracks remaining balance, calculates change, deducts ingredients, and updates `total_sales_revenue`.
Void checkLowResourceAlerts (void)	Emits operator warnings if water <= 30 L, soap <=200 mL, or wax <=50 mL.
void adminMode(void)	Authenticates passcode (`1357`). On success, runs an infinite admin loop offering inventory inspection, replenishment, price editing, and revenue resetting.
void displayInventoryAndSales (void)	Prints formatted current supply levels and total sales revenue.
void replenishIngredients(void)	Prompts operator to enter add quantities for water, soap, and wax. Updates inventory levels without exceeding maximum tank capacities.
void changePackagePrices (void)	Allows the operator to select any wash package and update its unit price in AED.
void resetSalesRevenue(void)	Prompts confirmation, reminds operator to collect physical cash box revenue, and resets sales counter to 0.00 AED.

6. Assessment Tasks & Grading Rubric

Assessment Component	Evaluation Criteria	Marks
Pre-Lab Design Tasks	Accurate Function Hierarchy Chart and Payment Flowchart.	5
Main Architecture & Menus	Proper `main()` loop, `#define` constants, and menu navigation.	5
Customer Mode & Payment Loop	Resource availability check, step-by-step coin insertion, coin validation, change calculation, and inventory deduction.	20
Admin Operations Mode	Passcode check (1357), user replenishment with max capacity checking, price edit, and revenue reset.	10
In-Lab Demonstration & QA	Demonstrate working code to lab instructor.	10
Lab progress	Complete weekly deliverables in class	20
Final Report	Complete report	30

Appendix A: Sample Terminal Run Transcripts

The sample terminal executions below illustrate the expected visual output and behaviour.

Sample Run 1: Customer Wash Purchase with Coin Payment

```
============================================================
   WELCOME TO THE AUTOMATED CAR WASH CONTROLLER SYSTEM
============================================================

------------------------------------------------------------
                    MAIN SYSTEM MENU
------------------------------------------------------------
  1. Purchase a Car Wash Package (Customer Mode)
  2. Maintenance & Operations (Admin Mode)
  3. Exit Application
------------------------------------------------------------
Enter your choice (1-3): 1

------------------------------------------------------------
               AVAILABLE CAR WASH PACKAGES
------------------------------------------------------------
  1. Basic Eco Wash       : AED 15.00
  2. Deluxe Foam Wash     : AED 25.00
  3. Premium Shine & Wax  : AED 40.00
  0. Cancel & Return to Main Menu
------------------------------------------------------------
Select a wash package (0-3): 2

You selected option 2. Confirm order? (1 = Yes, 0 = No/Re-select): 1
```

```
============================================================
 PAYMENT PROCESS: Deluxe Foam Wash
 TOTAL AMOUNT DUE: AED 25.00
 Accepted Coins  : 1.00 AED, 0.50 AED, 0.25 AED
============================================================

 Outstanding Balance : AED 25.00
 Insert Coin (1.0, 0.50, 0.25): 1
 [ACCEPTED] Coin AED 1.00 added. Total Paid: AED 1.00

 Outstanding Balance : AED 24.00
 Insert Coin (1.0, 0.50, 0.25): 1
 [ACCEPTED] Coin AED 1.00 added. Total Paid: AED 2.00

 Outstanding Balance : AED 23.00
 Insert Coin (1.0, 0.50, 0.25): 2
 [REJECTED] Invalid coin (AED 2.00). Please collect your coin.

 Outstanding Balance : AED 23.00
 Insert Coin (1.0, 0.50, 0.25): 1
 [ACCEPTED] Coin AED 1.00 added. Total Paid: AED 3.00
```

...

```
 Outstanding Balance : AED 1.00
 Insert Coin (1.0, 0.50, 0.25): 1
 [ACCEPTED] Coin AED 1.00 added. Total Paid: AED 25.00


============================================================
 SUCCESS! DISPENSING WASH SERVICE: Deluxe Foam Wash
------------------------------------------------------------
 Package Price : AED 25.00
 Total Paid    : AED 25.00
 Change Due    : AED 0.00
============================================================
 Please drive your vehicle into the bay. Thank you!

------------------------------------------------------------
                   MAIN SYSTEM MENU
------------------------------------------------------------
  1. Purchase a Car Wash Package (Customer Mode)
  2. Maintenance & Operations (Admin Mode)
  3. Exit Application
------------------------------------------------------------
Enter your choice (1-3):
```

Sample Run 2: Admin Operations & Manual Inventory Replenishment

```
------------------------------------------------------------
                    MAIN SYSTEM MENU
------------------------------------------------------------
  1. Purchase a Car Wash Package (Customer Mode)
  2. Maintenance & Operations (Admin Mode)
  3. Exit Application
------------------------------------------------------------
Enter your choice (1-3): 2

------------------------------------------------------------
              ADMIN MAINTENANCE MODE ACCESS
------------------------------------------------------------
Enter Admin Passcode: 1354

[ACCESS DENIED] Incorrect passcode entered!
```

```
------------------------------------------------------------
                    MAIN SYSTEM MENU
------------------------------------------------------------
  1. Purchase a Car Wash Package (Customer Mode)
  2. Maintenance & Operations (Admin Mode)
  3. Exit Application
------------------------------------------------------------
Enter your choice (1-3): 2

------------------------------------------------------------
              ADMIN MAINTENANCE MODE ACCESS
------------------------------------------------------------
Enter Admin Passcode: 1357

[ACCESS GRANTED] Welcome, System Operator.

------------------------------------------------------------
                ADMIN OPERATIONS MENU
------------------------------------------------------------
  1. Display Inventory Levels & Sales Revenue
  2. Replenish Ingredient Inventory
  3. Change Wash Package Prices
  4. Reset Total Sales Revenue
  0. Exit Admin Mode
------------------------------------------------------------
Select an admin operation (0-4): 2
```

```
------------------------------------------------------------
             REPLENISH INGREDIENT INVENTORY
------------------------------------------------------------
Current Water: 150.0 L (Max: 300.0 L)
Enter Water to add (Liters): 50
Current Soap: 800.0 mL (Max: 2000.0 mL)
Enter Soap to add (mL): 200
Current Wax: 400.0 mL (Max: 1000.0 mL)
Enter Wax to add (mL): 260

============================================================
 [SUCCESS] INVENTORY REPLENISHED SUCCESSFULLY!
------------------------------------------------------------
  Updated Water Level : 200.0 L
  Updated Soap Level  : 1000.0 mL
  Updated Wax Level   : 660.0 mL
============================================================

------------------------------------------------------------
                 ADMIN OPERATIONS MENU
------------------------------------------------------------
  1. Display Inventory Levels & Sales Revenue
  2. Replenish Ingredient Inventory
  3. Change Wash Package Prices
  4. Reset Total Sales Revenue
  0. Exit Admin Mode
------------------------------------------------------------
```

END OF LAB